

Chapter 2 — Field-Effect Transistors (FET)

Electronics Device and Circuit (ENEX 151 / EX 151) · Tribhuvan University, IOE · BE (BEI, BCT) · Year I / Part II

Exam-oriented notes with embedded Past-Year Questions (2081 Ashwin R · 2082 Baishakh B · 2082 Bhadra R · 2083 Baishakh B)

References: Boylestad-Nashelsky (bias circuits — all four PYQ numericals are from here), Sedra-Smith (device physics, K-convention), Floyd, Millman-Halkias

Figures: rendered locally from the governing device equations using the exam papers' own parameter values (web download unavailable in this environment — see the fetch report)

How to Use These Notes / Chapter Overview

A BJT is a *current*-controlled device: base current lets collector current flow, and both carrier types take part. A field-effect transistor instead controls the drain current with an electric **field** set up by the gate *voltage*: no steady input current, one carrier type only (**unipolar**), and an input resistance so large it is effectively an open circuit. The chapter is one connected story:

JFET (squeeze a built-in channel with reverse-biased junctions) → **E-MOSFET** (create a channel from nothing through an oxide) → its **I-V law** → **D-MOSFET** (a channel you can both squeeze and enrich) → **biasing** → **DC circuit analysis** → the **common-source amplifier** → **CMOS logic**.

CROSS-CHAPTER PYQ: 2083 Q1(a) (asked inside the Chapter-1 question, 1 mark): “Why is a FET called a unipolar device?” — Because conduction uses only ONE type of charge carrier (electrons in n-channel, holes in p-channel), flowing through a single doped channel. A BJT is bipolar: both electrons and holes participate. Keep this one-liner ready.

PYQ Frequency Analysis — FET is worth exactly 13 marks every single year

Topic	2081 Ashwin	2082 Baishakh	2082 Bhadra	2083 Baishakh	Verdict
Construction + working of one FET type (curves + mathematics)	✓ E-MOSFET [7]	✓ D-MOSFET [7]	✓ JFET [6]	✓ D-MOSFET [6]	★ 4/4 GUARANTEED — the device rotates
DC numerical: find I_D , V_{DS} (voltage-divider bias)	✓ JFET [6]	✓ JFET + g_m [6]	✓ E-MOSFET [7]	✓ JFET [7]	★ 4/4 GUARANTEED
Why FET is unipolar (definition)	-	-	-	✓ (inside Q1) [1]	1/4 — one-liner

Strategic conclusions

- ★ Prepare the full **construction + working + characteristic curves + equations** answer for **all three devices** (JFET, E-MOSFET, D-MOSFET). Rotation so far: E-MOS (2081) → D-MOS (2082 Ba) → JFET (2082 Bh) → D-MOS (2083). D-MOSFET has appeared twice — but all three remain live; the answer skeleton is identical, only the physics paragraph changes.
- ★ Master the **voltage-divider numerical** cold. The method never changes: Thévenin gate voltage → bias line $V_{GS} = V_G - I_D R_S$ → device law (Shockley or square law) → quadratic → **reject the impossible root** → V_{DS} → **saturation / pinch-off check**. All four papers, 6–7 marks, all four solved in §2.6 with every arithmetic step.
- All four PYQ circuits are lifted from Boylestad's worked examples, so the marking scheme expects Boylestad's method — shown here analytically, with the graphical cross-check in Fig. 8.

2.1 Structure and Physical Operation of the JFET

◆ Intuition first

Take a bar of n-type silicon — just a resistor with electrons drifting from Source to Drain. Now embed two p⁺ regions in its sides and reverse-bias those p–n junctions. Every reverse-biased junction grows a **depletion region** — an insulating zone emptied of carriers — which eats into the bar and narrows the conducting channel. The gate voltage therefore works like a tap: more reverse bias → wider depletion → thinner channel → less current. Because the controlling junction is always *reverse-biased*, essentially no current ever flows into the gate: the JFET is controlled by a field, not by a current.

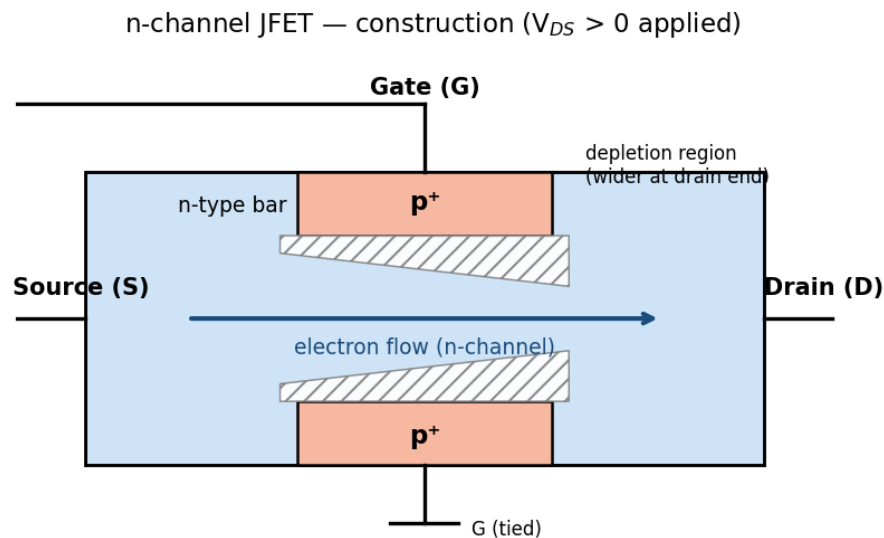


Fig. 1 — n-channel JFET construction. Note the depletion wedges are *WIDER* at the drain end — the exam examiner looks for this. [rendered locally]

Formal description of operation — the three-stage story (this is the marked structure of the answer)

- **Stage 1 — $V_{GS} = 0$, small V_{DS} :** the channel behaves as a simple resistor; I_D rises linearly with V_{DS} (**ohmic / triode region**). But the drain end of the channel sits at a higher potential than the source end, so the gate-channel reverse bias is larger near the drain — the depletion wedges are thicker there (Fig. 1).
- **Stage 2 — $V_{GS} = 0$, V_{DS} raised:** the wedges almost touch at the drain end when $V_{DS} = |V_p|$ — **pinch-off**. Current does NOT stop: carriers are injected into the pinched zone and swept through by its strong field. Extra V_{DS} is absorbed across the pinched zone, so I_D **saturates** at I_{DSS} (Drain-to-Source Saturation current — the maximum the device ever carries in normal operation).
- **Stage 3 — V_{GS} made negative:** uniform extra squeezing. The channel starts narrower, pinches off earlier (at $V_{DS} = V_{GS} - V_p$) and saturates at a lower current. At $V_{GS} = V_p$ (the pinch-off / cut-off voltage, e.g. -4 V) the depletion regions close the channel completely: $I_D = 0$ (**cutoff**).

JFET characteristics — plotted from $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ with $I_{DSS}=8\text{ mA}$, $V_P=-4\text{ V}$ (2083 Q4 values)

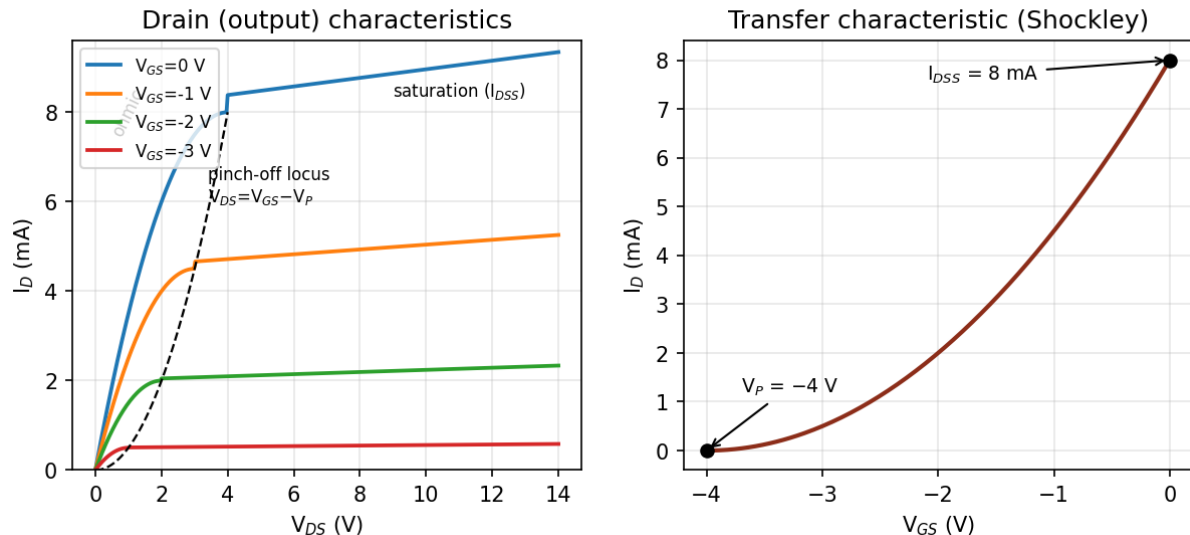


Fig. 2 — JFET output and transfer characteristics, computed from Shockley's equation with the 2083 Q4 device values ($I_{DSS} = 8\text{ mA}$, $V_P = -4\text{ V}$). [rendered from equations]

Equations (the “mathematical expressions” every PYQ demands)

Shockley's equation (saturation region): $I_D = I_{DSS} (1 - V_{GS}/V_P)^2$ valid for $V_P \leq V_{GS} \leq 0$ and $V_{DS} \geq V_{GS} - V_P$

Symbol	Meaning	Notes
I_{DSS}	Maximum drain current, measured at $V_{GS} = 0$	datasheet value
V_P ($\equiv V_{GS(off)}$)	Pinch-off voltage — NEGATIVE for n-channel	$I_D = 0$ when $V_{GS} = V_P$
Ohmic-region law	$I_D \approx I_{DSS} [2(1 - V_{GS}/V_P)(V_{DS}/ V_P) - (V_{DS}/ V_P)^2]$	mention only; boundary at $V_{DS} = V_{GS} - V_P$

Derivation — transconductance g_m (differentiating Shockley, 2 lines)

$g_m \equiv dI_D/dV_{GS} = I_{DSS} \cdot 2(1 - V_{GS}/V_P) \cdot (-1/V_P) = (2I_{DSS}/|V_P|)(1 - V_{GS}/V_P)$. Defining the maximum value at $V_{GS} = 0$:

$$g_{m0} = 2I_{DSS}/|V_P| \quad g_m = g_{m0}(1 - V_{GS}/V_P) = (2/|V_P|) \cdot \sqrt{I_{DSS} \cdot I_D}$$

Input resistance: the gate junction is reverse-biased, so I_G is a leakage of nano-amps $\rightarrow R_{in} \approx 10^8\text{--}10^{10}\ \Omega$. This is why in every bias circuit we treat the gate as drawing **zero current** — the fact that makes FET voltage-divider analysis exact rather than approximate.

JFET vs BJT — the comparison the theory question often closes with

Property	JFET	BJT
Control	voltage (V_{GS}), field effect	current (I_B), carrier injection
Carriers	one type — unipolar	both — bipolar
Input resistance	$\approx 10^8\text{--}10^{10}\ \Omega$	$\approx 10^3\ \Omega$ ($\beta \cdot r_e$)
Transfer law	square law (Shockley)	exponential ($I_C = I_S e^{V_{BE}/V_T}$)
g_m at same current	lower	higher (I_C/V_T)
Noise / thermal stability	lower noise, no thermal runaway	higher noise, runaway possible

Exam interpretation + $\triangle$ common mistakes

- PYQ 2082 Bhadra Q3 [6]:** “Describe the construction and working principle of n-channel JFET with the help of characteristic curves and mathematical expressions.” $\rightarrow$ exactly this section: Fig. 1 (≈ 1.5 marks) + the three-stage story (≈ 2.5) + Fig. 2 curves with regions labelled (≈ 1) + Shockley and g_m

boxes (≈ 1). ● **HIGH**

- ⚠ Writing $V_p = +4$ V inside formulas — it is **negative** for n-channel; only $|V_p|$ appears in g_{m0} .
- ⚠ Drawing uniform depletion regions — they must be **wider at the drain end** whenever $V_{DS} > 0$.
- ⚠ Saying current stops at pinch-off — it **saturates**; pinch-off is where the ohmic region ends, not where conduction ends.

■ **Section summary:** reverse-biased junction depletion squeezes a built-in n-channel; ohmic → pinch-off → saturation; Shockley's parabola from $(V_p, 0)$ to $(0, I_{DSS})$; g_m grows linearly toward g_{m0} as $V_{GS} \rightarrow 0$; gate draws no current.

2.2 Structure and Physical Operation of the Enhancement-Type MOSFET

◆ Intuition first

The JFET can only *narrow* a channel that already exists. The enhancement MOSFET goes further: it has **no channel at all** until the gate *creates* one. The trick is the Metal-Oxide-Semiconductor sandwich: the gate is a capacitor plate separated from the silicon by an insulating SiO_2 film. Charge the plate positively and it pulls electrons up to the silicon surface; enough of them and the surface locally turns n-type — a conducting bridge appears between two n^+ islands. The gate literally manufactures the wire.

n-channel enhancement MOSFET — construction (no built-in channel)

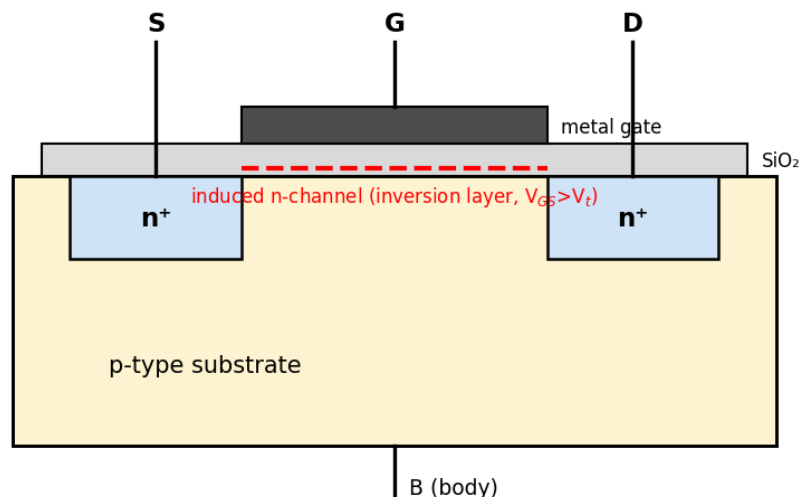


Fig. 3 — n-channel E-MOSFET construction. There is NO built-in channel — the dashed line only appears when V_{GS} exceeds V_t . [rendered locally]

Physical operation — the four-step story (PYQ 2081 Ashwin Q4 [7])

- **Step 1 — $V_{GS} = 0$:** drain-to-source is two back-to-back p-n junctions; whatever the polarity of V_{DS} , one of them is reverse-biased $\rightarrow I_D \approx 0$. The device is **normally OFF**.
- **Step 2 — small positive V_{GS} :** the gate-oxide-substrate capacitor charges. Positive gate charge first pushes the substrate's holes away from the surface, leaving a depletion layer.
- **Step 3 — $V_{GS} \geq V_t$ (threshold voltage, typically 1-3 V):** the field is strong enough to attract minority electrons to the surface until they outnumber holes there — an **inversion layer**: an induced n-channel joining the two n^+ wells. Conduction begins; raising V_{GS} further deepens the channel and lowers its resistance — the channel is **enhanced**, hence the name. The excess $V_{OV} = V_{GS} - V_t$ is the **overdrive voltage**.
- **Step 4 — increasing V_{DS} :** the channel is shallowest at the drain end (local gate-to-channel voltage is smaller there). At $V_{DS} = V_{GS} - V_t$ the drain end pinches off $\rightarrow I_D$ **saturates** at a value fixed by V_{GS} . Same saturation idea as the JFET, new mechanism for creating the channel.

Input resistance: the gate is separated by an insulator, so gate current is exactly zero (not merely leakage-small); $R_{in} \approx 10^{10}$ - $10^{15} \Omega$ — even higher than the JFET. (Practical corollary: MOS gates are static-sensitive; protective zeners are built in.)

EXAM: Answer template for 2081 Q4 [7]: Fig. 3 with all four terminals labelled (≈ 2 marks) + the four steps above (≈ 3) + the drain characteristic curve and square-law equations of §2.3, Fig. 4 (≈ 2). Sections 2.2 and 2.3 together ARE the 7-mark answer.

2.3 Current-Voltage Characteristics of the Enhancement MOSFET

Where the square law comes from (derivation sketch worth 1-2 bonus marks)

The induced channel is a capacitor plate: its mobile charge per unit area is $Q = C_{ox}(V_{GS} - V_t)$. Current = charge $\times$ velocity, and velocity = $\mu_n E = \mu_n V_{DS}/L$. Multiplying and accounting for the average channel voltage gives the **triode law**; at the saturation boundary $V_{DS} = V_{OV}$ it collapses to the **square law**. With $K \equiv \frac{1}{2} \mu_n C_{ox} (W/L)$ — exactly the K given in PYQ 2082 Bhadra Q4:

$$\text{Triode } (V_{DS} < V_{GS} - V_t): I_D = K[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$$

$$\text{Saturation } (V_{DS} \geq V_{GS} - V_t): I_D = K(V_{GS} - V_t)^2$$

Symbol	Meaning	Notes
$V_t (\equiv V_{GS(th)})$	Threshold voltage	no channel below it
$K \text{ (mA/V}^2\text{)}$	Conduction parameter = $\frac{1}{2} k'_n (W/L) = \frac{1}{2} \mu_n C_{ox} (W/L)$	Boylestad's datasheet form: $K = I_{D(on)} / (V_{GS(on)} - V_t)^2$ — state the equivalence if asked
Channel-length modulation	saturation curves tilt slightly: $\times(1 + \lambda V_{DS})$	$r_o = 1/(\lambda I_D) \approx V_A/I_D$
$g_m \text{ (saturation)}$	$2K(V_{GS} - V_t) = 2\sqrt{K I_D}$	differentiate the square law

E-MOSFET characteristics — $I_D = K(V_{GS} - V_t)^2$ with $K=0.48 \text{ mA/V}^2$, $V_t=1 \text{ V}$ (2082 Bhadra Q4 values)

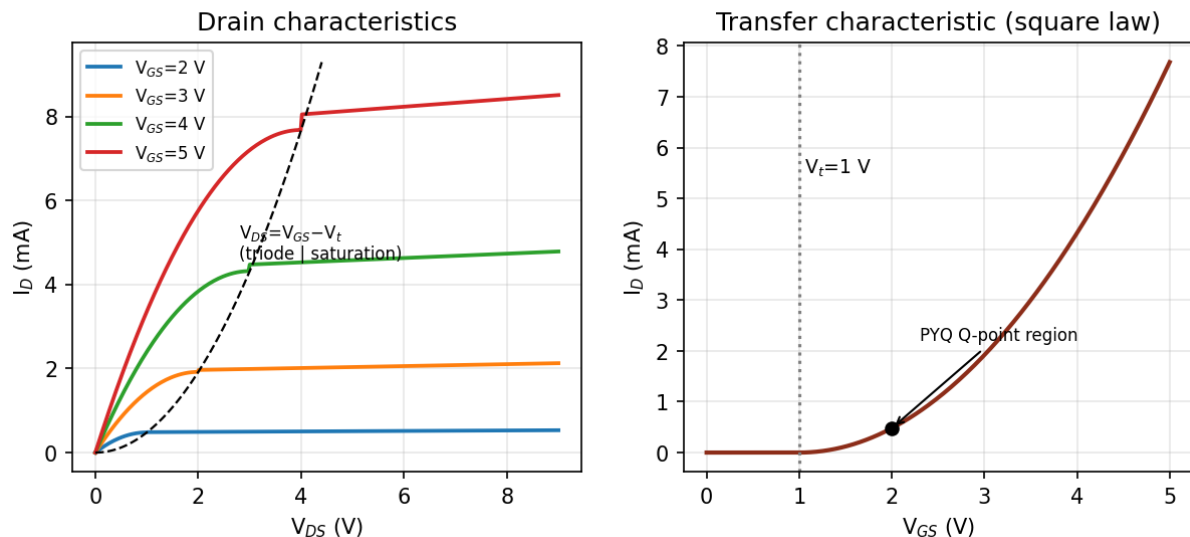


Fig. 4 — E-MOSFET characteristics computed from the square law with the 2082 Bhadra Q4 device ($K = 0.48 \text{ mA/V}^2$, $V_t = 1 \text{ V}$). Dashed parabola = triode/saturation boundary $V_{DS} = V_{GS} - V_t$. [rendered from equations]

- $\triangle$ The boundary is $V_{DS} = V_{GS} - V_t$, NOT $V_{DS} = V_t$.
- $\triangle$ Transfer curve starts at V_t (zero below it) — do not draw it through the origin.

2.4 The Depletion-Type MOSFET ★ REPEATED PYQ TOPIC

Asked twice in four papers — 2082 Baishakh Q4 [7] and 2083 Baishakh Q3 [6], both worded “describe the construction and working principle of an n-channel depletion-type MOSFET with characteristic curves and mathematical expressions.”

◆ Intuition

Take the E-MOSFET and diffuse a real, physical n-channel between the two n^+ wells at manufacture. Now current flows even at $V_{GS} = 0$ (normally ON, like a JFET) — but because the gate is insulated, you may drive it to **either polarity**: negative gate voltage empties (depletes) the channel, positive gate voltage enriches (enhances) it. It is the only device of the three that works on both sides of $V_{GS} = 0$.

n-channel depletion MOSFET — construction (channel diffused in at manufacture)

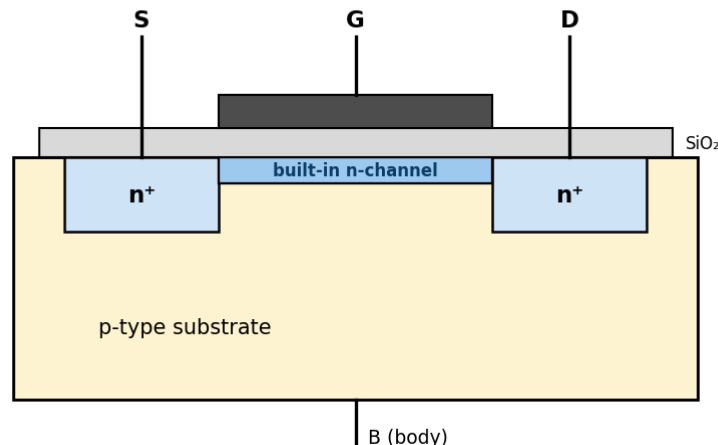


Fig. 5 — D-MOSFET construction — identical to Fig. 3 EXCEPT the thin built-in n-channel. State this one-line difference explicitly: it is the first mark. [rendered locally]

Working principle — the two modes (the core of the 6-7 mark answer)

- **Depletion mode ($V_{GS} < 0$):** the negative gate charge induces positive charge in the channel — i.e. it repels electrons out of it. Channel conductivity falls, $I_D < I_{DSS}$; at $V_{GS} = V_p$ the channel is fully emptied → cutoff. Behaviour mirrors the JFET, but the mechanism is electrostatic through the oxide, not junction depletion.
- **Enhancement mode ($V_{GS} > 0$):** positive gate charge attracts *additional* electrons into the channel: I_D rises **above** I_{DSS} . A JFET can never do this — its gate junction would forward-bias and destroy the high input resistance. This half of the story is what distinguishes your answer from a JFET answer: never omit it.
- **Effect of V_{DS} :** same as before — ohmic region, pinch-off at $V_{DS} = V_{GS} - V_p$, saturation.

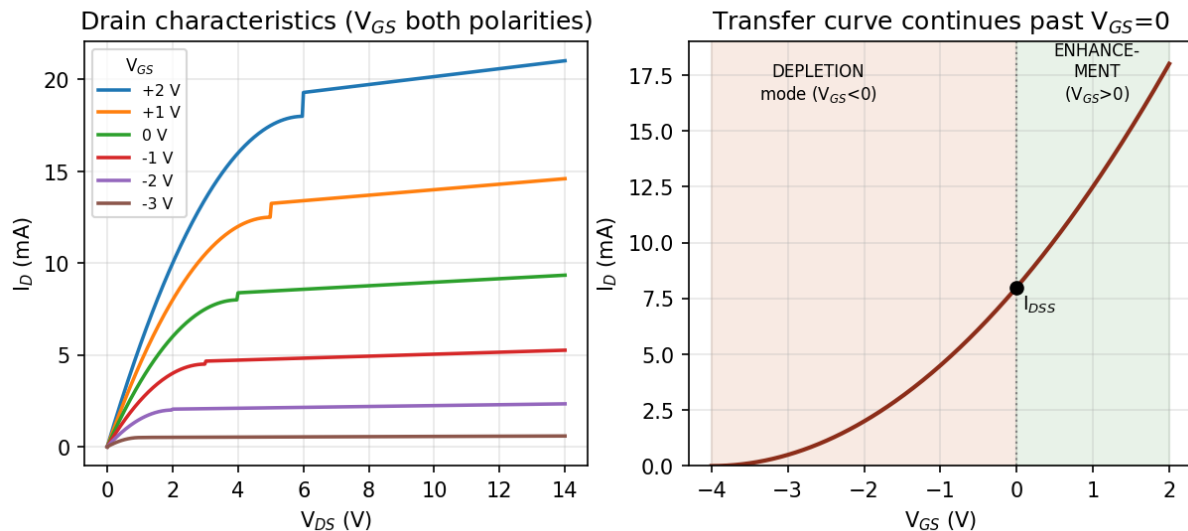


Fig. 6 — D-MOSFET characteristics: drain curves exist for BOTH polarities of V_{GS} , and the transfer parabola continues past $V_{GS} = 0$ above I_{DSS} . [rendered from equations]

Mathematics

Same Shockley equation governs BOTH modes: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ (V_{GS} now allowed > 0) • $g_m = g_{m0}(1 - V_{GS}/V_P)$, $g_{m0} = 2I_{DSS}/|V_P|$

- ⚠ Do NOT use the E-MOSFET square law $K(V_{GS} - V_t)^2$ here — the D-MOSFET follows Shockley.
- ⚠ Symbols: D-MOSFET symbol has a **solid** channel bar, E-MOSFET a **broken** one — drawing the wrong symbol costs the construction mark.
- ⚠ Forgetting the enhancement half of the transfer curve = answering about a JFET; this is the single most common lost mark.

■ **Section summary + PYQ check:** both PYQs (2083 Q3, 2082 Ba Q4) = Fig. 5 + the two-mode explanation + Fig. 6 + the formula box, closing with one sentence contrasting it against JFET (junction vs oxide) and E-MOS (built-in vs induced channel). ● **HIGH**

2.5 Biasing in MOS / FET Amplifier Circuits

◆ The biasing problem

An amplifier needs a stable quiescent point $Q = (I_{DQ}, V_{DSQ})$ in the saturation region. For a JFET/D-MOSFET the gate must normally sit *below* the source ($V_{GS} \leq 0$); for an E-MOSFET it must sit *above* it by more than V_t . Every practical circuit is a way of generating that offset without wasting the FET's infinite input resistance.

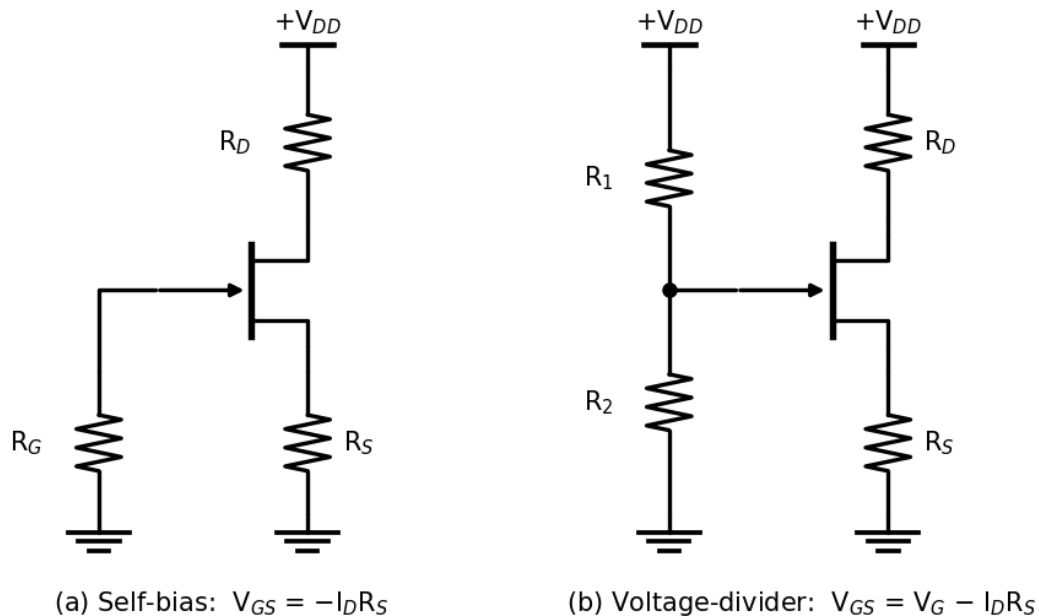


Fig. 7 — The two JFET bias workhorses. (a) Self-bias: the source resistor generates its own negative V_{GS} . (b) Voltage-divider bias — the skeleton of ALL FOUR PYQ numericals. [rendered locally]

- **Fixed bias:** a separate negative supply V_{GG} drives the gate: $V_{GS} = -V_{GG}$. Simple, but needs a second supply and gives no stabilisation — listed first only to be rejected.
- **Self-bias:** gate grounded through R_G (no gate current $\rightarrow V_G = 0$); the drain current lifts the source: $V_{GS} = 0 - I_D R_S = -I_D R_S$. Negative feedback: if I_D drifts up, V_{GS} goes more negative and pulls it back.
- **Voltage-divider bias (the exam circuit):** R_1 - R_2 fix $V_G = V_{DD} R_2 / (R_1 + R_2)$ exactly (zero gate current!); then $V_{GS} = V_G - I_D R_S$. Works for JFET ($I_D R_S$ deliberately exceeds V_G , making V_{GS} negative) and for E-MOSFET (divider set above V_t). Strongest stabilisation; M Ω -range resistors keep Z_{in} huge.
- **Drain-feedback bias (E-MOSFET only):** R_G from drain to gate; with no gate current $V_{GS} = V_{DS}$ — guarantees saturation since $V_{DS} = V_{GS} > V_{GS} - V_t$ automatically.

Master equations of the divider circuit: $V_G = V_{DD} \cdot R_2 / (R_1 + R_2)$ · $V_{GS} = V_G - I_D R_S$ · $V_{DS} = V_{DD} - I_D (R_D + R_S)$

2.6 FET / MOSFET Circuits at DC — the guaranteed 6-7 mark numerical ★ REPEATED PYQ TOPIC

◆ **Universal analytical procedure (write these steps in the exam — they carry method marks)**

Step	Action	Why it earns a mark
1	$V_G = V_{DD} R_2 / (R_1 + R_2)$	valid because $I_G = 0$ — say so
2	Bias line: $V_{GS} = V_G - I_D R_S$	links circuit to device

Step	Action	Why it earns a mark
3	Substitute into Shockley (JFET/D-MOS) or $K(V_{GS}-V_t)^2$ (E-MOS, assume saturation)	device law
4	Solve the quadratic; REJECT the root violating $V_p < V_{GS} \leq 0$ (JFET) or $V_{GS} > V_t$ (E-MOS)	stating WHY the root is rejected is itself worth a mark
5	$V_{DS} = V_{DD} - I_D(R_D + R_S)$	output side
6	Verify region: $V_{DS} \geq V_{GS} - V_p$ (JFET) or $\geq V_{GS} - V_t$ (E-MOS)	confirms step 3's assumption — examiners ask for it explicitly (2081 Q5)
7	g_m if asked: $g_{m0}(1 - V_{GS}/V_p)$ or $2K(V_{GS}-V_t)$	small-signal bridge

★ **Worked PYQ 1 — 2083 Baishakh Q4 [7] (JFET; Boylestad's classic 2.1 M Ω / 270 k Ω example)**

Given: $V_{DD} = 16$ V, $R_1 = 2.1$ M Ω , $R_2 = 270$ k Ω , $R_D = 2.4$ k Ω , $R_S = 1.5$ k Ω ; $V_p = -4$ V, $I_{DSS} = 8$ mA. Find I_D and V_{DS} .

- $V_G = 16 \times 270k / 2370k = 4320/2370 = \mathbf{1.82$ V
- $V_{GS} = 1.82 - 1.5 \cdot I_D$ (I_D in mA)
- $I_D = 8(1 + V_{GS}/4)^2 = (8/16)(4 + V_{GS})^2 = 0.5(4 + 1.82 - 1.5I_D)^2 = 0.5(5.82 - 1.5I_D)^2 = 0.5(33.91 - 17.47I_D + 2.25I_D^2)$
- $1.125I_D^2 - 9.73I_D + 16.95 = 0 \rightarrow I_D = [9.73 \pm \sqrt{(94.76 - 76.29)}/2.25 = (9.73 \pm 4.30)/2.25 \rightarrow \mathbf{2.42$ mA or 6.24 mA. Reject 6.24 mA: it gives $V_{GS} = 1.82 - 9.36 = -7.53$ V $< V_p = -4$ V — the device would be cut off, contradiction.
- Q-point:** $I_{DQ} \approx \mathbf{2.42$ mA, $V_{GSQ} = 1.82 - 3.63 = \mathbf{-1.80$ V ✓ (between V_p and 0)
- $V_{DS} = 16 - 2.42m \times (2.4k + 1.5k) = 16 - 9.44 = \mathbf{6.56$ V
- Check: $V_{GS} - V_p = -1.80 + 4 = 2.2$ V; $V_{DS} = 6.56$ V > 2.2 V ✓ saturation — Shockley was valid.

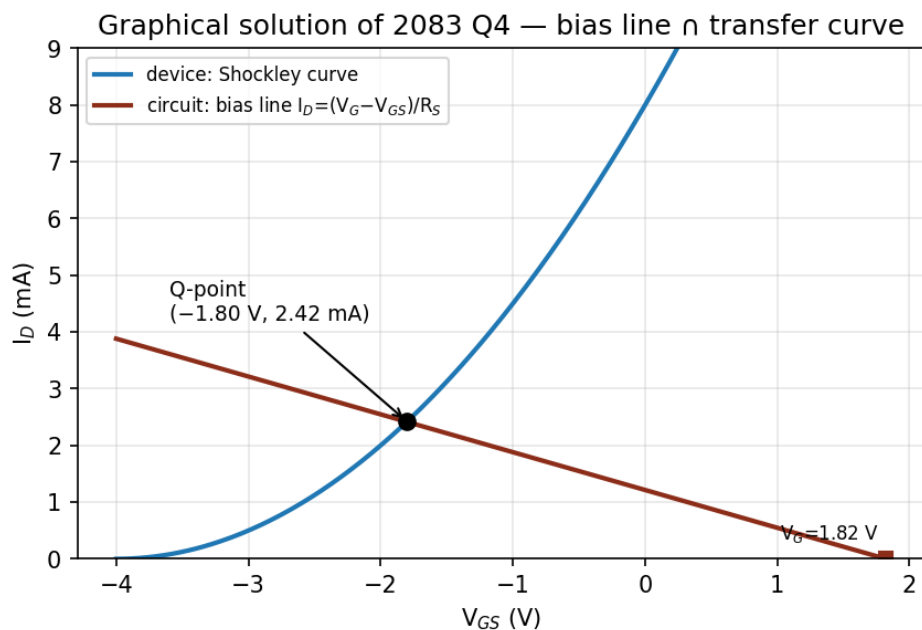


Fig. 8 — The same problem solved graphically (Boylestad's method): the circuit's bias line intersects the device's transfer curve at the Q-point. Use this as a sanity check for your quadratic. [rendered from equations]

★ **Worked PYQ 2 — 2082 Bhadra Q4 [7] (E-MOSFET)**

Given: $V_{DD} = 10$ V, $R_1 = R_2 = 11$ M Ω , $R_D = R_S = 6$ k Ω ; $V_t = 1$ V, $K = 0.48$ mA/V². Find I_D and V_{DS} .

- $V_G = 10 \times 11/22 = \mathbf{5$ V
- $V_{GS} = 5 - 6I_D$
- Assume saturation: $I_D = 0.48(V_{GS} - 1)^2 = 0.48(4 - 6I_D)^2 = 0.48(16 - 48I_D + 36I_D^2) = 7.68 - 23.04I_D + 17.28I_D^2$
- $17.28I_D^2 - 24.04I_D + 7.68 = 0 \rightarrow I_D = [24.04 \pm \sqrt{(577.9 - 530.8)}/34.56 = (24.04 \pm 6.86)/34.56 \rightarrow \mathbf{0.50$ mA or 0.89 mA. Reject 0.89 mA: $V_{GS} = 5 - 5.37 = -0.37$ V $< V_t$ — device would be OFF yet

carrying current, contradiction.

- **Q-point:** $I_D \approx 0.5 \text{ mA}$, $V_{GS} = 5 - 3.0 = 2.0 \text{ V} \checkmark (> V_t)$
- **5.** $V_{DS} = 10 - 0.5 \text{ mA} \times 12 \text{ k} = 4.0 \text{ V}$ **6.** $V_{OV} = 1.0 \text{ V}$; $4.0 > 1.0 \checkmark$ saturation confirmed.

★ **Worked PYQ 3 — 2082 Baishakh Q5 [6] (JFET + g_m ; Boylestad's 910 k / 110 k example)**

Given: $V_{DD} = 20 \text{ V}$, $R_1 = 910 \text{ k}\Omega$, $R_2 = 110 \text{ k}\Omega$, $R_D = 2.2 \text{ k}\Omega$, $R_S = 1.1 \text{ k}\Omega$; $I_{DSS} = 10 \text{ mA}$, $V_p = -3.5 \text{ V}$. Find I_{DQ} , V_{GSQ} , V_{DS} and g_m .

- **1.** $V_G = 20 \times 110/1020 = 2.16 \text{ V}$ **2.** $V_{GS} = 2.16 - 1.1 \cdot I_D$
- **3.** $I_D = 10(1 + V_{GS}/3.5)^2 = (10/12.25)(3.5 + 2.16 - 1.1I_D)^2 = 0.816(5.66 - 1.1I_D)^2 = 26.12 - 10.16I_D + 0.988I_D^2$
- **4.** $0.988I_D^2 - 11.16I_D + 26.12 = 0 \rightarrow I_D = (11.16 \pm \sqrt{21.31})/1.976 = (11.16 \pm 4.62)/1.976 \rightarrow 3.31 \text{ mA}$ or 7.99 mA . Reject 7.99 mA ($V_{GS} = -6.63 \text{ V} < V_p$).
- **Q-point:** $I_{DQ} \approx 3.31 \text{ mA}$, $V_{GSQ} = 2.16 - 3.64 = -1.49 \text{ V}$
- **5.** $V_{DS} = 20 - 3.31 \text{ mA} \times 3.3 \text{ k} = 20 - 10.92 = 9.08 \text{ V}$ ($V_S = 3.64 \text{ V}$, $V_D = 12.7 \text{ V}$ if asked)
- **7.** $g_{m0} = 2 \times 10 \text{ mA}/3.5 = 5.71 \text{ mS} \rightarrow g_m = 5.71 \text{ m}(1 - 1.49/3.5) = 5.71 \text{ m} \times 0.574 = 3.28 \text{ mS}$

★ **Worked PYQ 4 — 2081 Ashwin Q5 [6] (JFET + explicit pinch-off check)**

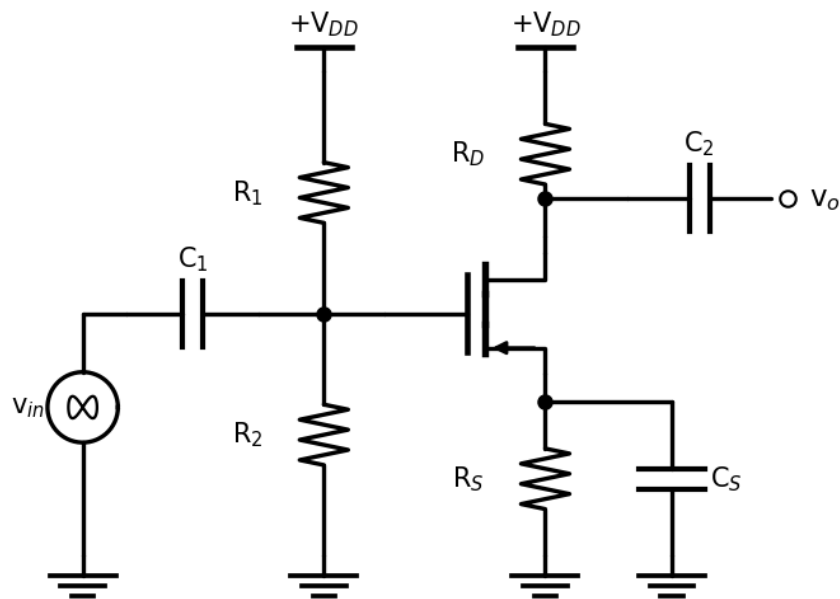
Given: $V_{DD} = 20 \text{ V}$, $R_1 = 1.5 \text{ M}\Omega$, $R_2 = 330 \text{ k}\Omega$, $R_D = 2 \text{ k}\Omega$, $R_S = 1 \text{ k}\Omega$; $V_p = -5.5 \text{ V}$, $I_{DSS} = 10 \text{ mA}$. Capacitors ideal. Find I_D , V_{DS} , and check whether the transistor operates in the pinch-off region.

- **1.** $V_G = 20 \times 330/1830 = 3.61 \text{ V}$ **2.** $V_{GS} = 3.61 - 1 \cdot I_D$
- **3.** $I_D = 10(1 + V_{GS}/5.5)^2 = (10/30.25)(5.5 + 3.61 - I_D)^2 = 0.331(9.11 - I_D)^2 = 27.42 - 6.02I_D + 0.331I_D^2$
- **4.** $0.331I_D^2 - 7.02I_D + 27.42 = 0 \rightarrow I_D = (7.02 \pm \sqrt{13.04})/0.661 = (7.02 \pm 3.61)/0.661 \rightarrow 5.16 \text{ mA}$ or 16.1 mA . Reject 16.1 mA : it exceeds I_{DSS} and gives $V_{GS} < V_p$.
- **Q-point:** $I_{DQ} \approx 5.16 \text{ mA}$, $V_{GSQ} = 3.61 - 5.16 = -1.55 \text{ V}$
- **5.** $V_{DS} = 20 - 5.16 \text{ mA} \times 3 \text{ k} = 20 - 15.47 = 4.53 \text{ V}$
- **6. The asked check:** pinch-off requires $V_{DS} \geq V_{GS} - V_p = -1.55 + 5.5 = 3.95 \text{ V}$. Since $4.53 \text{ V} > 3.95 \text{ V} \rightarrow \text{YES, the JFET operates in the pinch-off (saturation) region} \checkmark$ — write this sentence; it is the final mark.

EXAM TRICK: The wrong quadratic root ALWAYS violates the device constraint — check it in one line, never agonise. Sanity band: $0 < I_{DQ} < I_{DSS}$ for a JFET; $V_{GS} > V_t$ for an E-MOSFET. If your V_{DS} comes out negative, you kept the wrong root.

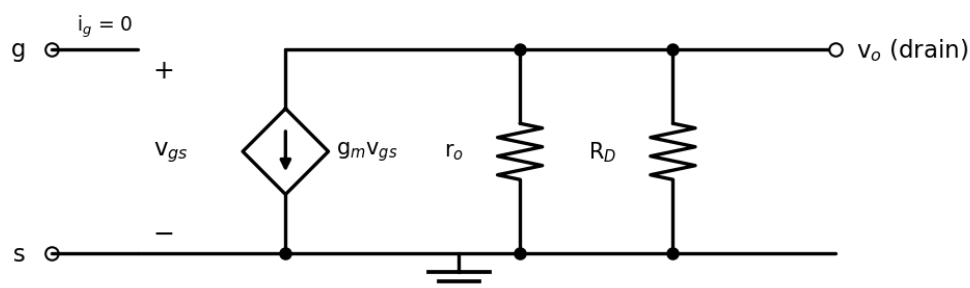
2.7 The MOSFET as an Amplifier — the Common-Source Stage

Bias the device at Q in saturation (everything of §2.5–2.6), superimpose a small signal on the gate, and the square law linearises to $i_d = g_m v_{gs}$. The small-signal model is minimal: an open circuit at the gate and a dependent current source at the drain.



Common-source amplifier (voltage-divider bias, bypassed R_S)

Fig. 9 — Common-source amplifier: voltage-divider bias, input/output coupling capacitors, and bypassed source resistor. [rendered locally]



$$A_v = v_o/v_{gs} = -g_m(r_o \parallel R_D)$$

Fig. 10 — Small-signal equivalent circuit of the CS stage (capacitors shorted, V_{DD} grounded for AC). [rendered locally]

Derivations (each is two lines — reproduce them, don't memorise blindly)

- **Input impedance:** the gate is open $\rightarrow$ only the bias resistors load the source: $Z_i = R_1 \parallel R_2$ (or R_G) — M Ω -range, the FET's selling point.
- **Output impedance:** kill $v_{in} \rightarrow v_{gs} = 0 \rightarrow$ the current source dies $\rightarrow$ looking back: $Z_o = R_D \parallel r_o \approx R_D$.
- **Voltage gain:** $v_o = -g_m v_{gs}(R_D \parallel r_o)$ and $v_{gs} = v_{in} \rightarrow A_v = -g_m(R_D \parallel r_o \parallel R_L)$. The minus sign = 180° phase inversion.
- **Unbypassed R_S (swamped stage):** series feedback divides the gain: $A_v = -g_m R_D / (1 + g_m R_S)$ — lower but more stable/linear.

Worked mini-example (re-using PYQ values — how a fresh exam question would look)

Take the 2082 Baishakh Q5 circuit as an amplifier: $g_m = 3.28 \text{ mS}$, $R_D = 2.2 \text{ k}\Omega$, $R_1 \parallel R_2 = 910\text{k} \parallel 110\text{k} = 98.1 \text{ k}\Omega$; take $r_o = 40 \text{ k}\Omega$. Then $Z_i = \mathbf{98.1 \text{ k}\Omega}$; $Z_o = 2.2\text{k} \parallel 40\text{k} = \mathbf{2.09 \text{ k}\Omega}$; $A_v = -3.28\text{m} \times 2.09\text{k} = \mathbf{-6.8}$ (-7.2 if r_o neglected). Note the modest gain versus a BJT CE stage — FET g_m is smaller at equal current; the FET buys impedance, not gain.

● **MEDIUM** — no FET-amplifier PYQ has appeared yet in your four papers (the BJT versions were asked instead). It is an untouched syllabus item and therefore a live candidate for a fresh 4–6 mark question; prepare the three derivations above.

2.8 MOSFET and CMOS as Logic Circuits

An NMOS inverter (driver transistor + load) pulls its output low when driven, but conducts static current in that state and needs a compromise load. The fix pairs complementary devices:

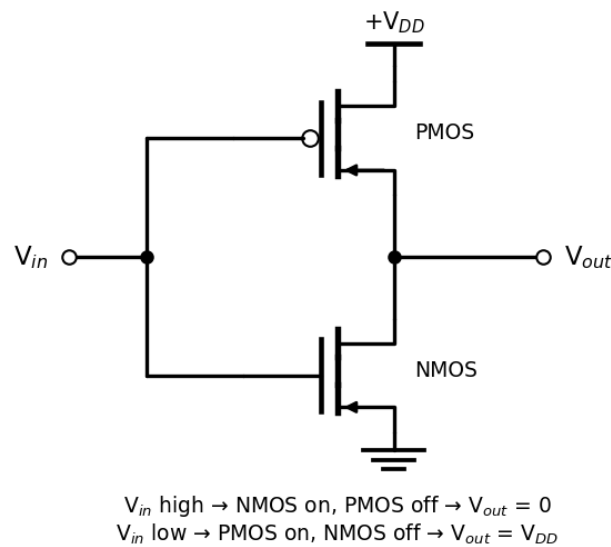


Fig. 11 — The CMOS inverter — in *EITHER* logic state one transistor is off, so the static supply current is essentially zero. [rendered locally]

- $V_{in} = \text{HIGH} \rightarrow \text{NMOS on (pulls output to 0), PMOS off. } V_{in} = \text{LOW} \rightarrow \text{PMOS on (pulls output to } V_{DD}), \text{ NMOS off.}$
- **Only one device ever conducts in steady state $\rightarrow \approx$ zero static power;** energy is spent only while switching ($P \approx f \cdot C \cdot V_{DD}^2$). Add rail-to-rail output swing, huge noise margins, and enormous packing density — and you have why CMOS owns digital electronics.

● **LOW** — never asked in your papers; hold a 2–4 mark readiness (operation + the zero-static-power argument).

FET Final Revision

Conceptual summary (read this the night before)

A JFET squeezes a built-in channel using reverse-biased junction depletion: normally ON, $V_{GS} \leq 0$, Shockley's parabola between $(V_p, 0)$ and $(0, I_{DSS})$, pinch-off then saturation. An E-MOSFET has no channel until $V_{GS} > V_t$ induces an inversion layer through the oxide: normally OFF, square law $K(V_{GS} - V_t)^2$. The D-MOSFET carries a built-in channel *and* an insulated gate, so Shockley's law extends to both polarities — depletion below zero, enhancement above. All three are unipolar, voltage-controlled, with near-infinite input resistance. Bias them with a divider plus source resistor; solve DC by intersecting the bias line with the device law (quadratic → reject the impossible root), finish with V_{DS} and the saturation check. As an amplifier the CS stage gives $A_v = -g_m(R_D \parallel r_o)$ with MΩ input impedance; as logic, complementary MOSFETs give CMOS and its zero static power.

Important diagrams checklist

Figure	Priority	Marked in PYQ as
Fig. 1 — JFET construction (depletion wedges wider at drain)	● HIGH	2082 Bhadra Q3
Fig. 2 — JFET drain + transfer curves	● HIGH	every JFET theory Q
Fig. 3 — E-MOSFET construction (no channel!)	● HIGH	2081 Q4
Fig. 4 — E-MOS curves + triode/saturation boundary	● HIGH	2081 Q4
Fig. 5 — D-MOSFET construction (built-in channel)	● HIGH	2083 Q3, 2082 Ba Q4 ★
Fig. 6 — D-MOS both-modes curves	● HIGH	2083 Q3, 2082 Ba Q4 ★
Fig. 7 — self-bias and divider-bias circuits	● HIGH	all four numericals
Fig. 8 — bias line n transfer curve (graphical Q)	● MEDIUM	cross-check method
Fig. 9/10 — CS amplifier + small-signal model	● MEDIUM	fresh-question candidate
Fig. 11 — CMOS inverter	● LOW	never asked

Formula sheet

Shockley: $I_D = I_{DSS}(1 - V_{GS}/V_p)^2$ · $g_{m0} = 2I_{DSS}/|V_p|$ · $g_m = g_{m0}(1 - V_{GS}/V_p)$

Square law: $I_D = K(V_{GS} - V_t)^2$ (sat) · $K[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$ (triode) · $g_m = 2K(V_{GS} - V_t) = 2\sqrt{K \cdot I_D}$

Bias/DC: $V_G = V_{DD}R_2/(R_1 + R_2)$ · $V_{GS} = V_G - I_D R_S$ · $V_{DS} = V_{DD} - I_D(R_D + R_S)$ · **checks:** $V_{DS} \geq V_{GS} - V_p$ / $V_{GS} - V_t$

CS amp: $Z_i = R_1 \parallel R_2$ · $Z_o = R_D \parallel r_o$ · $A_v = -g_m(R_D \parallel r_o)$ · **unbypassed R_S :** $A_v = -g_m R_D / (1 + g_m R_S)$ · $r_o = 1/(\lambda I_D)$

High-priority topics / common mistakes / numerical types

- **HIGH** Device construction + working ×3 (★ 4/4 papers) · divider-bias numerical with root rejection and region check (★ 4/4).
- **MEDIUM** g_m at the Q-point · CS amplifier derivations · self-bias variant ($V_G = 0$ special case).
- **LOW** CMOS logic · ohmic-region law · drain-feedback bias.
- ⚠ Top mistakes: sign of V_p · accepting the wrong quadratic root · skipping the saturation check · square law applied to a D-MOSFET · forgetting $I_G = 0$ makes the divider exact · E-MOSFET drawn with a built-in channel · transfer curve through the origin.
- Numerical types: **N1** divider-bias JFET (PYQs 1, 3, 4) · **N2** divider-bias E-MOS (PYQ 2) · **N3** g_m at Q (PYQ 3) · **N4** self-bias variant — same method with $V_G = 0$.

Complete FET PYQ Bank (wording preserved)

Year / Exam	Q	Question (condensed, wording preserved)	Mark s	Type
2083 Baishakh (Back)	3	Describe the construction and working principle of N-Channel D-MOSFET with the help of characteristic curves and mathematical expressions.	6	Theory
2083 Baishakh (Back)	4	Find I_D and V_{DS} for the given circuit. Given: $V_P = -4$ V, $I_{DSS} = 8$ mA. (16 V; 2.1 M Ω /270 k Ω ; 2.4 k Ω ; 1.5 k Ω)	7	Numerical
2082 Bhadra (Regular)	3	Describe the construction and working principle of n-channel JFET with the help of characteristic curves and mathematical expressions.	6	Theory
2082 Bhadra (Regular)	4	Find I_D and V_{DS} for the circuit shown. Given: $V_t = 1$ V, $K = 0.48$ mA/V ² . (10 V; 11 M Ω /11 M Ω ; 6 k Ω ; 6 k Ω)	7	Numerical
2082 Baishakh (Back)	4	Describe the construction and working principle of n-channel depletion type MOSFET with the help of characteristics curve and mathematical expressions.	7	Theory
2082 Baishakh (Back)	5	For the given JFET voltage divider configuration, determine I_{DQ} , V_{GSQ} , V_{DS} and transconductance g_m . (20 V; 910 k Ω /110 k Ω ; 2.2 k Ω ; 1.1 k Ω ; $I_{DSS} = 10$ mA, $V_P = -3.5$ V)	6	Numerical
2081 Ashwin (Regular)	4	Describe the construction and working principle of n-channel enhancement-type MOSFET with the help of necessary diagrams and its drain characteristic curve.	7	Theory
2081 Ashwin (Regular)	5	Find I_D and V_{DS} for the following circuit. Given $V_P = -5.5$ V, $I_{DSS} = 10$ mA. Assume capacitors ideal and check whether the transistor is operating in the Pinch-off region or not. (20 V; 1.5 M Ω /330 k Ω ; 2 k Ω ; 1 k Ω)	6	Numerical

PYQ → What to Study Mapping

PYQ	Topic	Notes section(s)	Required skill
2083 Q3 [6]	D-MOSFET theory	§2.4 (Figs. 5–6); contrast §2.1–2.3	construction difference + two modes + curves + Shockley/ g_m
2083 Q4 [7]	JFET numerical	§2.5, §2.6 PYQ 1 (Fig. 8)	divider → quadratic → root rejection → V_{DS} → check
2082 Bh Q3 [6]	JFET theory	§2.1 (Figs. 1–2)	3-stage operation + curves + Shockley + g_m
2082 Bh Q4 [7]	E-MOSFET numerical	§2.3, §2.6 PYQ 2	square-law quadratic + $V_{GS} > V_t$ root + saturation check
2082 Ba Q4 [7]	D-MOSFET theory	§2.4	as 2083 Q3; at 7 marks add symbol + both transfer curves
2082 Ba Q5 [6]	JFET + g_m	§2.6 PYQ 3; g_m from §2.1	full DC solution + g_{m0} , g_m
2081 Q4 [7]	E-MOSFET theory	§2.2 + §2.3 (Figs. 3–4)	inversion-layer story + drain curves + square law
2081 Q5 [6]	JFET + pinch-off check	§2.6 PYQ 4	DC solution + explicit $V_{DS} \geq V_{GS} - V_P$ verdict

Final Coverage Audit

- ✓ Syllabus 2.1–2.8 all present, in order; 2.7 and 2.8 covered at readiness depth although never asked (coverage rule).
- ✓ References honoured: Boylestad (all four PYQ circuits and the analytic/graphical method, g_m forms), Sedra-Smith (device physics, $K = \frac{1}{2}k'_n W/L$ convention, channel-length modulation, CMOS), Floyd/Millman corroboration. Notation unified and flagged: $V_P \equiv V_{GS(off)}$, $V_t \equiv V_{GS(th)}$.
- ✓ All 8 FET PYQs embedded in their sections AND repeated in the bank + mapping; all four numerals solved with every arithmetic step, root rejection justified, and region checks written out.

- ✓ ★ Repeated topics flagged (device theory 4/4, DC numerical 4/4); ● priorities assigned from PYQ evidence; 11 figures rendered locally from the governing equations using the exam papers' own parameter values (web image download blocked in this environment — see `figs/fetch_report` note in the title block).

End of Chapter 2 notes. Companion documents: Chapter 1 (BJT) markdown notes and Chapters 3–5 compact notes; this PDF supersedes the compact Chapter-2 markdown.